Database Management System LAB
Spring Semester 2024-25
Lab Assignment 3

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Ans – 1) Foreign keys are to be added later on in the tables.

Note that in all the tables below Initial order is not the same as in Assignment because of addition of Primary Keys, the rows are by DEFAULT sorted in order of PRIMARY KEY.

```
mysql> create table Employee
    -> (Fname Varchar(20),
    -> Minit Char(1),
    -> Lname Varchar(20),
    -> Ssn int PRIMARY KEY,
    -> Bdate DATE,
    -> Address Varchar(50),
    -> Sex Char(1),
    -> Salary int,
    -> Super_ssn int,
    -> Dno int);
```

```
mysql> Insert into Employee
    -> Values
Query OK, 1 row affected (0.02 sec)9,'1965-01-09','731 Fondren, Houston, TX','M',30000,333445555,5);

mysql> Insert into Employee
    -> Values
    -> ('Franklin','T','Wong',333445555,'1955-12-08','638 Voss,Houston, TX','M',40000,888665555,5),
    -> ('Alicia','J','Zelaya',999887777,'1968-01-19','3321 Castle, Spring, TX','F',25000,987654321,4),
    -> ('Jennifer','S','Wallace',987654321,'1941-06-20','291 Berry, Bellaire, TX','F',43000,888995555,4),
    -> ('Ramesh','K','Narayan',666884444,'1962-09-15','975 Fire Oak, Humble, TX','M',38000,333445555,5),
    -> ('Joyce','A','English',453453453,'1972-07-31','5631 Rice, Houston, TX','F',25000,333445555,5),
    -> ('Ahmad','V','Jabbar',987987987,'1969-03-29','980 Dallas, Houston, TX','M',25000,987654321,4),
    -> ('James','E','Borg',888665555,'1937-11-10','450 Stone, Houston, TX','M',55000,NULL,1);
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> select * from Employee;
+----------+-------+---------+-----------+------------+-------------------------+------+--------+-----------+------+
| Fname    | Minit | Lname   | Ssn       | Bdate      | Address                 | Sex  | Salary | Super_ssn | Dno  |
+----------+-------+---------+-----------+------------+-------------------------+------+--------+-----------+------+
| John     | B     | Smith   | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M    |  30000 | 333445555 |    5 |
| Franklin | T     | Wong    | 333445555 | 1955-12-08 | 638 Voss, Houston, TX   | M    |  40000 | 888665555 |    5 |
| Joyce    | A     | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX  | F    |  25000 | 333445555 |    5 |
| Ramesh   | K     | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M    |  38000 | 333445555 |    5 |
| James    | E     | Borg    | 888665555 | 1937-11-10 | 450 Stone, Houston, TX  | M    |  55000 |      NULL |    1 |
| Jennifer | S     | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX | F    |  43000 | 888665555 |    4 |
| Ahmad    | V     | Jabbar  | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX | M    |  25000 | 987654321 |    4 |
| Alicia   | J     | Zelaya  | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX | F    |  25000 | 987654321 |    4 |
+----------+-------+---------+-----------+------------+-------------------------+------+--------+-----------+------+
8 rows in set (0.00 sec)
```

```
mysql> Create table Department
    -> (Dname Varchar(20),
    ->  Dnumber int PRIMARY KEY,
    -> Mgr_ssn int,
    -> Mgr_start_date DATE);
Query OK, 0 rows affected (0.05 sec)

mysql> insert into Department
    -> Values
    -> ('Research',5,333445555,'1988-05-22'),
    -> ('Administration',4,987654321,'1995-01-01'),
    -> ('Headquarters',1,888665555,'1981-06-19');
Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from Department;
+----------------+---------+-----------+----------------+
| Dname          | Dnumber | Mgr_ssn   | Mgr_start_date |
+----------------+---------+-----------+----------------+
| Headquarters   |       1 | 888665555 | 1981-06-19     |
| Administration |       4 | 987654321 | 1995-01-01     |
| Research       |       5 | 333445555 | 1988-05-22     |
+----------------+---------+-----------+----------------+
3 rows in set (0.00 sec)
```

```
mysql> create table Dept_Locations
    -> (Dnumber int,
    -> DLocation Varchar(20));
Query OK, 0 rows affected (0.10 sec)

mysql> insert into Dept_Locations
    -> Values
    -> (1,'Houston'),
    -> (4,'Stafford'),
    -> (5,'Bellaire'),
    -> (5,'Sugarland'),
    -> (5,'Houston');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from Dept_Locations;
+---------+-----------+
| Dnumber | DLocation |
+---------+-----------+
|       1 | Houston   |
|       4 | Stafford  |
|       5 | Bellaire  |
|       5 | Sugarland |
|       5 | Houston   |
+---------+-----------+
5 rows in set (0.00 sec)
```

```
mysql> create table Project
    -> (Pname Varchar(20),
    -> Pnumber int PRIMARY KEY,
    -> Plocation Varchar(20),
    -> Dnum int);
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> insert into Project
    -> Values
    -> ('ProductX',1,'Bellaire',5),
    -> ('ProductY',2,'Sugarland',5),
    -> ('ProductZ',3,'Houston',5),
    -> ('Computerization',10,'Stafford',4),
    -> ('Reorganization',20,'Houston',1),
    -> ('Newbenefits',30,'Stafford',4);
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> select * from Project;
+-----------------+---------+-----------+------+
| Pname           | Pnumber | Plocation | Dnum |
+-----------------+---------+-----------+------+
| ProductX        |       1 | Bellaire  |    5 |
| ProductY        |       2 | Sugarland |    5 |
| ProductZ        |       3 | Houston   |    5 |
| Computerization |      10 | Stafford  |    4 |
| Reorganization  |      20 | Houston   |    1 |
| Newbenefits     |      30 | Stafford  |    4 |
+-----------------+---------+-----------+------+
6 rows in set (0.00 sec)
```

```
mysql> create table Works_on
    -> (Essn int,
    -> Pno int,
    -> Hours float);
Query OK, 0 rows affected (0.04 sec)

mysql> insert into Works_on
    -> Values
    -> (123456789,1,32.5),
    -> (123456789,2,7.5),
    -> (666884444,3,40.0),
    -> (453453453,1,20.0),
    -> (453453453,2,20.0),
    -> (333445555,2,10.0),
    -> (333445555,3,10.0),
    -> (333445555,10,10.0),
    -> (333445555,20,10.0),
    -> (999887777,30,30.0),
    -> (999887777,10,10.0),
    -> (987987987,10,35.0),
    -> (987987987,30,5.0),
    -> (987654321,30,20.0),
    -> (987654321,20,15.0),
    -> (888665555,20,NULL);
Query OK, 16 rows affected (0.09 sec)
Records: 16  Duplicates: 0  Warnings: 0

mysql> select * from Works_on;
+-----------+------+-------+
| Essn      | Pno  | Hours |
+-----------+------+-------+
| 123456789 |    1 |  32.5 |
| 123456789 |    2 |   7.5 |
| 666884444 |    3 |    40 |
| 453453453 |    1 |    20 |
| 453453453 |    2 |    20 |
| 333445555 |    2 |    10 |
| 333445555 |    3 |    10 |
| 333445555 |   10 |    10 |
| 333445555 |   20 |    10 |
| 999887777 |   30 |    30 |
| 999887777 |   10 |    10 |
| 987987987 |   10 |    35 |
| 987987987 |   30 |     5 |
| 987654321 |   30 |    20 |
| 987654321 |   20 |    15 |
| 888665555 |   20 |  NULL |
+-----------+------+-------+
16 rows in set (0.00 sec)
```

```
mysql> create table Dependent
    -> (Essn int,
    -> Dependent_name Varchar(20),
    -> Sex Char(1),
    -> Bdate DATE,
    -> Relationship Varchar(20));
Query OK, 0 rows affected (0.10 sec)

mysql> insert into Dependent
    -> Values
    -> (333445555,'Alice','F','1986-04-05','Daughter'),
    -> (333445555,'Theodore','M','1983-10-25','Son'),
    -> (333445555,'Joy','F','1958-05-03','Spouse'),
    -> (987654321,'Abner','M','1942-02-28','Spouse'),
    -> (123456789,'Michael','M','1988-01-04','Son'),
    -> (123456789,'Alice','F','1988-12-30','Daughter'),
    -> (123456789,'Elizabeth','F','1967-05-05','Spouse');
Query OK, 7 rows affected (0.06 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> select * from Dependent;
+-----------+----------------+------+------------+--------------+
| Essn      | Dependent_name | Sex  | Bdate      | Relationship |
+-----------+----------------+------+------------+--------------+
| 333445555 | Alice          | F    | 1986-04-05 | Daughter     |
| 333445555 | Theodore       | M    | 1983-10-25 | Son          |
| 333445555 | Joy            | F    | 1958-05-03 | Spouse       |
| 987654321 | Abner          | M    | 1942-02-28 | Spouse       |
| 123456789 | Michael        | M    | 1988-01-04 | Son          |
| 123456789 | Alice          | F    | 1988-12-30 | Daughter     |
| 123456789 | Elizabeth      | F    | 1967-05-05 | Spouse       |
+-----------+----------------+------+------------+--------------+
7 rows in set (0.00 sec)
```

Adding Foreign Keys

```
mysql> alter table Employee add Constraint Foreign Key (Dno) References Department(Dnumber);
Query OK, 8 rows affected (0.17 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> alter table Dept_Locations add Constraint Foreign Key (Dnumber) References Department(Dnumber);
Query OK, 5 rows affected (0.11 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> alter table Project add Constraint Foreign Key (Dnum) References Department(Dnumber);
Query OK, 6 rows affected (0.12 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> alter table Works_on add Constraint Foreign Key (Pno) References Project(Pnumber);
Query OK, 16 rows affected (0.12 sec)
Records: 16  Duplicates: 0  Warnings: 0

mysql> alter table Works_on add Constraint Foreign Key (Essn) References Employee(Ssn);
Query OK, 16 rows affected (0.17 sec)
Records: 16  Duplicates: 0  Warnings: 0

mysql> alter table Dependent add Constraint Foreign Key (Essn) References Employee(Ssn);
Query OK, 7 rows affected (0.17 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

a)

```
mysql> select p.Pname,d.Dname,Concat(e.Fname," ",e.Minit,". ",e.Lname) as Name,e.Address,e.Bdate
    -> from Project as p
    -> Join Department as d on p.Plocation = 'Stafford' and d.Dnumber = p.Dnum
    -> Join Employee as e on e.Ssn = d.Mgr_ssn and d.Dnumber = e.Dno;
+-----------------+----------------+---------------------+-------------------------+------------+
| Pname           | Dname          | Name                | Address                 | Bdate      |
+-----------------+----------------+---------------------+-------------------------+------------+
| Computerization | Administration | Jennifer S. Wallace | 291 Berry, Bellaire, TX | 1941-06-20 |
| Newbenefits     | Administration | Jennifer S. Wallace | 291 Berry, Bellaire, TX | 1941-06-20 |
+-----------------+----------------+---------------------+-------------------------+------------+
2 rows in set (0.00 sec)
```

b)

```
mysql> select Concat(e.Fname," ",e.Minit,". ",e.Lname) as Name,e.Salary*1.1 as Salary
    -> from Project as p
    -> Join Works_on as w on (p.Pname = 'ProductX' and p.Pnumber = w.Pno)
    -> Join Employee as e on w.Essn = e.Ssn;
+------------------+---------+
| Name             | Salary  |
+------------------+---------+
| John B. Smith    | 33000.0 |
| Joyce A. English | 27500.0 |
+------------------+---------+
2 rows in set (0.01 sec)
```

c)

```
mysql> select e.Fname,e.Minit,e.Lname,e.Dno,p.Pname
    -> from Works_on as w
    -> Join Employee as e on e.Ssn = w.Essn
    -> Join Project as p on p.Pnumber = w.Pno
    -> Order by e.Dno,e.Lname,e.Fname;
+----------+-------+---------+------+-----------------+
| Fname    | Minit | Lname   | Dno  | Pname           |
+----------+-------+---------+------+-----------------+
| James    | E     | Borg    |    1 | Reorganization  |
| Ahmad    | V     | Jabbar  |    4 | Computerization |
| Ahmad    | V     | Jabbar  |    4 | Newbenefits     |
| Jennifer | S     | Wallace |    4 | Reorganization  |
| Jennifer | S     | Wallace |    4 | Newbenefits     |
| Alicia   | J     | Zelaya  |    4 | Computerization |
| Alicia   | J     | Zelaya  |    4 | Newbenefits     |
| Joyce    | A     | English |    5 | ProductX        |
| Joyce    | A     | English |    5 | ProductY        |
| Ramesh   | K     | Narayan |    5 | ProductZ        |
| John     | B     | Smith   |    5 | ProductX        |
| John     | B     | Smith   |    5 | ProductY        |
| Franklin | T     | Wong    |    5 | ProductY        |
| Franklin | T     | Wong    |    5 | ProductZ        |
| Franklin | T     | Wong    |    5 | Computerization |
| Franklin | T     | Wong    |    5 | Reorganization  |
+----------+-------+---------+------+-----------------+
16 rows in set (0.00 sec)
```

d)

```
mysql> select Concat(e.Fname," ",e.Minit,". ",e.Lname) as Name,
    -> (select Concat(Fname," ",Minit,". ",Lname)
    -> from Employee
    -> where d.Dnumber = e.Dno and d.Mgr_ssn  = Ssn) as Manger_name
    -> from Employee as e
    -> Join Department as d on d.Dnumber = e.Dno
    -> Join Works_on as w on (w.Hours > 10.0 and w.Essn = e.Ssn)
    -> Join Project as p on (p.Pname = 'ProductX' and w.Pno = p.Pnumber);
+------------------+------------------+
| Name             | Manger_name      |
+------------------+------------------+
| John B. Smith    | Franklin T. Wong |
| Joyce A. English | Franklin T. Wong |
+------------------+------------------+
2 rows in set (0.02 sec)
```

e) Note that there Smith himself will always be included in the list since he is compared to himself. If required Smith can be left out by putting a check condition in the where clause

Essn != 123456789

```
mysql> select DISTINCT w.Essn from Works_on as w
    -> Join Works_on as w1 on (w1.Pno = w.Pno and w1.Hours = w.Hours)
    -> where w1.Essn = 123456789; -- The Ssn of John_Smith is 123456789 as given in qn (e)
+-----------+
| Essn      |
+-----------+
| 123456789 |
+-----------+
1 row in set (0.00 sec)
```

f)

```
mysql> select Concat(Fname," ",Minit,". ",Lname) as Name
    -> from Employee as e
    -> Join Dependent as D on (Essn = Ssn and Dependent_name = Fname and D.Sex = e.Sex);
Empty set (0.00 sec)
```

g)

```
mysql> select sum(e.Salary) as Sum_Salary,max(e.Salary) as max_Salary,min(e.Salary) as min_Salary,avg(e.Salary) as avg_Salary
    -> from Employee as e
    -> Join Department as d on e.Dno = d.Dnumber
    -> where d.Dname = 'Research';
+------------+------------+------------+------------+
| Sum_Salary | max_Salary | min_Salary | avg_Salary |
+------------+------------+------------+------------+
|     133000 |      40000 |      25000 | 33250.0000 |
+------------+------------+------------+------------+
1 row in set (0.06 sec)
```

h)

```
mysql> select Dno,COUNT(*) as More_than_30k
    -> from Employee
    -> where Salary > 30000
    -> Group by Dno;
+------+---------------+
| Dno  | More_than_30k |
+------+---------------+
|    1 |             1 |
|    4 |             1 |
|    5 |             2 |
+------+---------------+
3 rows in set (0.00 sec)
```

i)

```
mysql> update Employee
    -> set Dno = (Select Dnumber from Dept_Locations where Dlocation = 'Stafford')
    -> where YEAR(Bdate) = 1965;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from Employee;
+----------+-------+---------+-----------+------------+-------------------------+------+--------+-----------+------+
| Fname    | Minit | Lname   | Ssn       | Bdate      | Address                 | Sex  | Salary | Super_ssn | Dno  |
+----------+-------+---------+-----------+------------+-------------------------+------+--------+-----------+------+
| John     | B     | Smith   | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M    |  30000 | 333445555 |    4 |
| Franklin | T     | Wong    | 333445555 | 1955-12-08 | 638 Voss, Houston, TX    | M    |  40000 | 888665555 |    5 |
| Joyce    | A     | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX   | F    |  25000 | 333445555 |    5 |
| Ramesh   | K     | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M    |  38000 | 333445555 |    5 |
| James    | E     | Borg    | 888665555 | 1937-11-10 | 450 Stone, Houston, TX   | M    |  55000 |      NULL |    1 |
| Jennifer | S     | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX  | F    |  43000 | 888665555 |    4 |
| Ahmad    | V     | Jabbar  | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX  | M    |  25000 | 987654321 |    4 |
| Alicia   | J     | Zelaya  | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX  | F    |  25000 | 987654321 |    4 |
+----------+-------+---------+-----------+------------+-------------------------+------+--------+-----------+------+
8 rows in set (0.00 sec)
```

j) Similar to (e) note that Smith himself will always be included in the list and if required he can be excluded by putting a where clause **“ Lname != Smith ”** .

```
mysql> select Concat(Fname," ",Minit,". ",Lname) as Name
    -> from Employee
    -> Join Works_on on Essn = Ssn
    -> where (Select COUNT(*) from
    -> ((Select Pno from Works_on where Essn = (Select Ssn from Employee where Lname = 'Smith'))
    -> EXCEPT
    -> (Select Pno from Works_on where Essn = Ssn))
    -> as EX) = 0
    -> Group by Ssn,Name;
+------------------+
| Name             |
+------------------+
| John B. Smith    |
| Joyce A. English |
+------------------+
2 rows in set (0.00 sec)
```

k)

```
mysql> alter table Employee
    -> Add Column Commission int Default NULL;
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> update Employee
    -> Set Commission = 10000 where Sex = 'F';
Query OK, 3 rows affected (0.02 sec)
Rows matched: 3  Changed: 3  Warnings: 0
```

```
mysql> select Concat(Fname," ",Minit,". ",Lname) as Name,IFNULL(Salary+Commission,Salary) as Total_Salary
    -> from Employee;
+---------------------+--------------+
| Name                | Total_Salary |
+---------------------+--------------+
| John B. Smith       |        30000 |
| Franklin T. Wong    |        40000 |
| Joyce A. English    |        35000 |
| Ramesh K. Narayan   |        38000 |
| James E. Borg       |        55000 |
| Jennifer S. Wallace |        53000 |
| Ahmad V. Jabbar     |        25000 |
| Alicia J. Zelaya    |        35000 |
+---------------------+--------------+
8 rows in set (0.01 sec)
```

l) First we have to ensure the Foreign Key supports – **ON DELETE CASCADE**.

Since in the initial declaration of foreign key, **ON DELETE CASCADE** was not added we are going to add it manually now.

```
mysql> alter table Works_on
    -> Drop Foreign Key Works_on_ibfk_1;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> alter table Works_on
    -> Add Constraint Foreign Key (Pno)
    -> References Project(Pnumber) ON DELETE CASCADE;
Query OK, 16 rows affected (0.08 sec)
Records: 16  Duplicates: 0  Warnings: 0

mysql> Delete from Project
    -> where Pname = 'Newbenefits';
Query OK, 1 row affected (0.03 sec)
```

Notice that on Removing ‘Newbenefits’ Pnumber = 30, Its references from Child table **Works_on** also were automatically removed.

```
mysql> select * from Project; select * from Works_on;
+-----------------+---------+-----------+------+
| Pname           | Pnumber | Plocation | Dnum |
+-----------------+---------+-----------+------+
| ProductX        |       1 | Bellaire  |    5 |
| ProductY        |       2 | Sugarland |    5 |
| ProductZ        |       3 | Houston   |    5 |
| Computerization |      10 | Stafford  |    4 |
| Reorganization  |      20 | Houston   |    1 |
+-----------------+---------+-----------+------+
5 rows in set (0.00 sec)

+-----------+------+-------+
| Essn      | Pno  | Hours |
+-----------+------+-------+
| 123456789 |    1 |  32.5 |
| 123456789 |    2 |   7.5 |
| 666884444 |    3 |    40 |
| 453453453 |    1 |    20 |
| 453453453 |    2 |    20 |
| 333445555 |    2 |    10 |
| 333445555 |    3 |    10 |
| 333445555 |   10 |    10 |
| 333445555 |   20 |    10 |
| 999887777 |   10 |    10 |
| 987987987 |   10 |    35 |
| 987654321 |   20 |    15 |
| 888665555 |   20 |  NULL |
+-----------+------+-------+
13 rows in set (0.00 sec)
```

m)

```
mysql> select TimeStampDiff(YEAR,Bdate,CurDate()) as Age,Salary
    -> from Employee
    -> where Dno = 5;
+------+--------+
| Age  | Salary |
+------+--------+
|   69 |  40000 |
|   52 |  25000 |
|   62 |  38000 |
+------+--------+
3 rows in set (0.01 sec)
```